Okafor Lab Health and Safety Rules

Revision 5, effective 1 July 2026. Written by Dr. Tomasz Brenner, Lab Safety Officer; reviewed with Westbrook Environmental Health & Safety (EHS). These rules add to, and never replace, Westbrook EHS policy and your EHS training. If a rule here seems to conflict with EHS guidance, follow the stricter one and tell Tomasz.

Emergency contacts (also posted by every lab door)

Fire, serious injury, serious exposure, anyone unconscious	**911** (from any phone)
Westbrook University Police (non-emergency; after-hours EHS spill response)	555-0142
Westbrook EHS (weekdays 8:00–17:00)	555-0160
Occupational Health Clinic (weekdays; needlesticks and exposures)	555-0175
After hours: Westbrook Medical Center Emergency Department	911 or walk in
Facilities, 24-hour line (power, flooding, building problems)	555-0150
Tomasz Brenner, Lab Safety Officer	x4415, mobile 555-0127
Prof. Adaeze Okafor, PI	x4410, mobile 555-0110

1. Personal protective equipment (PPE) by room

In every lab room (412, 413, 415, 416): no food, drink, gum or cosmetics; closed-toe shoes and long trousers or equivalent; long hair tied back. Take gloves off before touching door handles, lift buttons, phones or keyboards in shared spaces.

Room	Minimum PPE
412 Protein biochemistry	White lab coat, nitrile gloves and safety glasses at the bench at all times.
412B Sonicator enclosure	As 412, plus hearing protection while the sonicator is running.
413 Tissue culture (BSL-2)	Blue lab coat, which stays in 413; nitrile gloves; safety glasses, or a face shield where there is a splash risk.
415 Instrument room	Lab coat and gloves when handling samples; safety glasses when handling chemicals. No gloves on the keyboards.
416 Freezers, dewars, cold room	Lab coat. Insulated gloves for -80 °C racks and dry ice. Face shield *and* cryogenic gloves for dispensing liquid nitrogen.
417 Desks / 418 Offices	No PPE. No chemicals or samples. 417 is the only room in the suite where food and drink are allowed.

2. BSL-2 practices (tissue culture room 413)

We work with human cell lines (such as HEK293 and HeLa) and primary cultures from mouse tissue, all at BSL-2. **The lab does not work with lentivirus or other viral vectors.** Any plan to do so requires an amended biosafety registration *before* any work starts: talk to Tomasz and Adaeze.

- Only people with a completed TC sign-off (Mei-Ling Zhou and Tomasz Brenner) may work in 413. Keep the door closed; the biohazard sign stays on it.
- All open manipulation of cells happens inside a biosafety cabinet. No mouth pipetting, ever.
- Avoid sharps. Never recap needles. Use plastic serological pipettes, not glass Pasteur pipettes, wherever possible.
- Aspirate media into the vacuum flask, which contains enough bleach to give at least 10% final bleach. The line has an in-line filter. Empty the flask when it is half full.
- Wipe down work surfaces with 70% ethanol before and after work. Use 10% bleach (made fresh daily in a dated bottle) for spills. After bleach, wipe steel surfaces with 70% ethanol or water to prevent corrosion.
- Centrifuge BSL-2 material only in sealed buckets or safety caps, and open them inside the cabinet.
- Move cultures between rooms in a closed, labelled secondary container.

- Wash your hands after removing gloves and before leaving 413.
- Working hours and the buddy rule are in Section 7. **No BSL-2 work alone after 8 pm.**

3. Chemical storage and incompatibilities

- **Flammable liquids** (ethanol, methanol, isopropanol, acetone): keep them in the flammables cabinet under the 412 fume hood, with no more than 1 L at a bench. Only the labelled flammable-safe refrigerator in 412 may hold flammables; ordinary fridges can ignite vapours.
- **Acids and bases** are kept in separate corrosives cabinets under the 412 sink (acids on the left, bases on the right), in secondary containment.
- **Oxidisers** (hydrogen peroxide, ammonium persulfate) are kept away from flammables and organic materials.
- Label every secondary container with the full chemical name (not only an abbreviation), the hazard, the date and your initials. Safety data sheets are available through the EHS website.
- **Fume hood required for:** phenol and chloroform, β-mercaptoethanol, formaldehyde, and weighing paraformaldehyde powder. Unpolymerised acrylamide is a neurotoxin: use precast gels where possible, otherwise handle it with gloves in the hood.

Never mix	Why
Bleach + acids	Releases chlorine gas.
Bleach + ammonia or ammonium salts	Releases chloramine gases.
Bleach + guanidinium thiocyanate (TRIzol-type reagents, many kit lysis and wash buffers)	Releases toxic gases, including hydrogen cyanide. **Never bleach these wastes**; collect them as chemical waste.
Sodium azide + acids; azide down the drain	Forms hydrazoic acid, and explosive metal azides in drain pipes. Collect azide-containing solutions as chemical waste.
Oxidisers + flammables or organics	Fire risk.

4. Waste streams

Waste	Where it goes
Solid biohazard (tips, flasks, gloves and plastics that touched cells)	Red biohazard bag in a lidded bin with the biohazard symbol. Autoclaved by the person on autoclave duty (see *Who Does What*), then placed in the building's treated-waste bin.
Liquid biohazard	Add bleach to 10% final and leave for at least 30 minutes, then pour down the sink with running water, **only if it contains no other hazardous chemicals**.
Sharps (needles, blades, glass pipettes, anything that can puncture), contaminated or not	Red sharps container. Close and lock it when three-quarters full. Never push waste down into it.
Broken glass, not contaminated	Cardboard glass box. Tape it shut when full and label it "Glass".
Chemical waste	The satellite accumulation area under the 412 fume hood. Use a compatible container, attach a hazardous waste tag at the first addition, and keep it closed. Keep separate: halogenated solvents; non-halogenated solvents; acids; bases; guanidinium waste; azide waste; acrylamide gels (solid chemical waste).
Mouse tissue and carcasses	*Never* regular or biohazard trash. They go back to the vivarium carcass freezer as the animal protocol requires.
Radioactive	**The lab has no radioactive materials authorisation. Radioactive materials may not be brought into or used in the lab.**
Clean paper and plastic	Regular trash; cardboard to recycling.

5. Spills and exposures

Biological spill inside a biosafety cabinet. Keep the cabinet running. Cover the spill with paper towels, gently apply 10% bleach from the edges inward, and leave it for 20 minutes. Discard everything as solid biohazard, then wipe all surfaces (including the grille area) with 70% ethanol. Tell Mei-Ling and Tomasz.

Biological spill outside a cabinet. Warn others and leave the room for 30 minutes so that aerosols settle. Close the door and post a sign. Remove any contaminated PPE. Come back in fresh PPE, cover the spill with paper towels, apply 10% bleach from the edges inward, wait 20 minutes, clean up as biohazard, and report it. If the spill is large, or you are unsure, call EHS (555-0160; after hours through University Police, 555-0142).

Chemical spill. For a small spill of a chemical you know and can handle safely, use a spill kit (412 by the door, 413 by the sink, the 416 corridor; the biological spill kit is in 413). Absorb, bag, and label it as chemical waste. **If the spill is large, of an unknown or highly toxic chemical, or anyone is hurt: evacuate, close the door, and call 911.** Do not try to clean it up.

Exposure to your body.

- *Skin:* remove contaminated clothing and rinse for 15 minutes under the safety shower or at a sink.
- *Eyes:* use the eyewash for 15 minutes, holding your eyelids open. Then seek medical attention.
- *Needlestick, cut or splash involving cells or other biological material:* wash with soap and water for 15 minutes (eyes: eyewash for 15 minutes). **Report it immediately** to Tomasz or Adaeze and go to Occupational Health (weekdays) or the Emergency Department (after hours) the same day. Do not wait until the next morning.

6. Emergency equipment locations

- **Eyewash stations:** 412 (at the sink by the door), 413 (at the sink, on your left as you enter the TC room), 415 (next to the sink). Flushed weekly as a rota chore.

- **Safety showers:** in 412 next to the fume hood, and in the corridor outside 413.
- **Fire extinguishers:** by every lab door. **Fire alarm pull station:** by the east stairwell. **First aid kits:** 414 and 413 (by the door). **AED:** fourth-floor lift lobby.

7. Working alone and after hours

"Working alone" means nobody else is within call in the lab suite (Keller 412–418).

- **A buddy is required at any hour** for: phenol/chloroform extractions; work with more than 1 L of flammable solvent; any procedure whose SOP says "buddy required".
- **No BSL-2 work alone after 8 pm**, on any day. On weekends and university holidays, no BSL-2 work alone at any time. Your buddy must be *physically present in the suite* and know what you are doing. **A Slack message or a text is not a buddy.**
- Dispensing liquid nitrogen from the supply dewar after 8 pm or on weekends also needs a buddy present.
- **Undergraduates and rotation students** do no bench work after 8 pm or on weekends unless a graduate student or postdoc from the lab is present in the suite.
- Low-risk work alone is fine at any hour: data analysis, reading, NanoDrop measurements, plate-reader or imager runs on prepared samples. After 8 pm, post your arrival and departure in #after-hours.
- Building access, dry ice, liquid nitrogen and freezer alarms are covered in *After Hours and Special Materials*.

8. Incident and near-miss reporting

Report every incident *and* near miss to Tomasz within 24 hours, and to Adaeze. Report injuries and exposures immediately. Injuries and exposures also need a Westbrook EHS Incident Report (online form, within 24 hours). Near misses go in the lab near-miss log (`okafor-nas/admin/safety/near_miss_log.xlsx`). We do not blame people for reporting. Unreported near misses are how serious accidents happen.

9. Inspections and records

Tomasz inspects the lab on the first Monday of each month, and EHS inspects it once a year. Biosafety cabinets are certified every year. Keep your training certificates up to date in `okafor-nas/admin/training/`. Bloodborne Pathogens training must be renewed every year.